

04_01: Master Data Extraction Table (The Technical Audit)

1. Purpose and Scope

This document serves as the primary "Dataset" for the systematic literature review titled: " **Datasets and Data Preprocessing Techniques for Machine Learning-Based Insider Threat Detection.**" It provides a study-by-study breakdown of the technical evidence synthesized from **110 primary studies** published between 2021 and 2025.

The objective of this table is to provide total transparency into how raw qualitative evidence from the literature was transformed into structured data to answer our **Four** core Research Questions (**RQ1–RQ4**).

2. Table Structure and Mapping

The spreadsheet is organized to follow the logical flow of a machine learning pipeline, from data sourcing to final architectural implementation. Each row corresponds to a unique primary study, with columns mapped as follows:

- **Study Metadata:** Unique ID, Author, and Year.
- **RQ1 (Dataset Characteristics):** Identifies the dataset (e.g., CERT r4.2, LANL), log modalities (e.g., logon, file access, email), and temporal duration.
- **RQ2 (Dataset Applicability):** Documents source type (synthetic vs. real-world), labeling quality, and inherent domain-specific limitations.
- **RQ3 (Preprocessing Pipeline):** Details the granular preprocessing steps (e.g., aggregation, missing-value imputation, normalization, and class imbalance handling strategies).
- **RQ4 (Data-Centric Challenges):** Documents evidence regarding reproducibility, ground truth gaps, privacy restrictions, and dataset heterogeneity.

3. Hybrid AI-Human Verification Protocol

To ensure the highest degree of technical accuracy, this data was extracted using a multi-phase verification protocol:

- **Phase I: AI-Assisted Extraction:** Google Gemini parsed the full-text PDFs of all 110 studies to generate initial data points based on a predefined structured extraction schema.

- **Phase II: Manual Expert Verification:** Each data point was cross-referenced by both authors against the original source manuscript. Any discrepancies were flagged and corrected.
- **Phase III: Technical Audit & Cross-Checking:** High-precision parameters—such as normalization formulas, specific window sizes for temporal features, and class-balancing ratios—were manually audited to ensure they matched the exact figures reported in the source literature.

#	Author (Year)	RQ1: Datasets & Preprocessing	RQ2: Expert/Manual Indicators	RQ3: Automated Representation Learning	RQ4 (Reported Data Challenges)
1	Adun (2023)	Dataset: CERT Steps: Correlation-based feature removal (expunging) and SVM-based classification for data labeling prior to the fuzzy training phase.	Indicators: Mapping 6 specific log attributes (Date, User, Source, Action, etc.) to fuzzy sets to establish a threat "Presence/Absence" baseline.	Architecture: Hybrid SVM-ANFIS. The Neuro-Fuzzy layers use a Least Square Estimator to learn if-then rules that capture latent, unpredictable behavioral patterns.	• <i>Limited real-world data</i> • <i>Dependence on synthetic datasets</i> • <i>Dataset heterogeneity</i> • <i>Class imbalance</i> • <i>Weak ground truth</i>
2	Ahmad (2025)	Dataset: Real-world production logs (60-90 days) from an academic institution. Steps: Rule-based filtering for noise reduction; Label Encoding for categorical features; and Min-Max scaling/standardization via the OpenSearch integration plugin.	Indicators: 10 "High-Impact Use Cases" derived from MITRE ATT&CK and Verizon DBIR statistics . Examples: Geolocation anomalies (18% of breaches), Windows Event ID 4672 (Privilege Escalation), and Event ID 4625 (Failed Logins).	Architecture: Random Cut Forest (RCF) integrated into Wazuh SIEM. RCF builds dynamic sketches of data streams; it identifies latent patterns by measuring the displacement/change in model complexity caused by a new data point, detecting "low and slow" lateral movements.	

3	Ahmadi (2025)	Dataset: Anonymized logs and resource utilization data from distributed corporate environments. Steps: Preprocessing involves extracting contextual features (IP, Time, Device) and normalizing them into a risk-scoring format to handle disparate environmental shifts.	Indicators: A formal Risk-Scoring Model ($R = \sum w_i \cdot f_i(x)$). Expert-weighted factors include: 1. Login patterns, 2. Geolocation/IP reputation, and 3. Sensitivity of the requested resource.	Architecture: An autonomous framework using K-Means Clustering to establish "Identity Segments" (behavioral baselines). It utilizes Random Forest to extract latent patterns in access requests, allowing the system to dynamically adjust permissions and isolate "compromised identities" in real-time.	• Privacy and anonymization • Inconsistent preprocessing reporting • Cross-dataset generalization
4	Alabdulkreem (2022)	Dataset: Aggregated logs from 5 sources (Removable device, Email, HTTP, File, Logon). Steps: Noise removal through aggregation into a single homogeneous file; feature subset selection to reduce noise; and feature encoding for deep learning input.	Indicators: Specific expert-defined rules used for baselining: 1. Frequency of removable device usage during non-office hours, 2. Access to high-risk URLs (wikileaks.org), 3. Anomalous login timestamps.	Architecture: MWF-IDLDC Model (Metaheuristic with Weighted Fusion). It uses an ensemble of LSTM and GRU to capture temporal sequences and SAE for unsupervised feature representation. The Ant Lion Optimizer (ALO) automatically extracts the most effective hyperparameter set to minimize the classifier error rate.	
5	Ali (2025)	Dataset: 1,306 Army Insider Threat Hub cases (narrative text). Steps: Text extraction via Python; Chunking (dividing	Indicators: Leverages expert-evaluated narratives containing dispositional (personality) and situational (external	Architecture: BERT-based NLP and BERTopic for Zero-shot topic modeling. The model uses bidirectional context to	

		documents into 512-token segments to avoid truncation); and class balancing through random duplication (over-sampling) and deletion (under-sampling).	triggers) traits. Adjudicated threat levels (Negligible, Low, Medium, High) serve as the manual baseline.	extract latent semantic patterns from "Analyst Research" narratives, reassembling chunked predictions to identify the highest risk signals in unstructured text.	
6	Al-Mhiqani (2021)	Dataset: CERT r4.2. Steps: Log parsing (text strings to numeric), MaxAbsScaler normalization, and ADASYN (Adaptive Synthetic) sampling. ADASYN is used to solve the class imbalance by generating synthetic data points for the minority (malicious) class based on their "difficulty level" in learning.	Indicators: Leverages five user event categories (Logon/Logoff, Device, Email, HTTP, and File). The baseline is established by the expert-injected malicious instances in the r4.2 dataset, representing 3 distinct scenarios.	Architecture: AD-DNN (Integrated ADASYN and Deep Neural Network). The DNN uses 20 hidden layers to extract latent hierarchical features. By combining this with ADASYN, the model learns to identify high-variance characteristics of low-frequency threats that standard ML typically misses.	
7	Al-Mhiqani (2022)	Dataset: CERT r4.2. Steps: Aggregation of heterogeneous logs; Parsing engine to convert strings to numeric; and an objective Entropy-based weighting to determine the importance of evaluation criteria for model selection.	Indicators: Expert-defined feature set (Table 5) covering 5 domains: Logon/Logoff, Device usage, Email metadata, HTTP URLs, and File access. Specific "Misuse" signatures are learned from the known attack labels in the dataset.	Architecture: Intelligent Multilayer Framework. Layer 1 uses Entropy-VIKOR (MCDM) to mathematically select the best algorithms. Layer 2 is a Hybrid HITD: Random Forest (Parallel ensemble trees) handles known signatures (Misuse), while KNN (Similarity-based	

				clustering) extracts latent patterns to detect "Unknown" (Anomaly) threats.	
8	Almusawy (2024)	Dataset: CERT r4.2. Steps: Used NoSQL (MongoDB/JSON style) for horizontal scaling and rapid aggregation of 32 million events into 1,000 user profiles. Applied SMOTE to generate synthetic minority samples to prevent overfitting on the small user-count baseline.	Indicators: Four specific manual features: 1. Logon frequency outside work hours, 2. Count of blocked website visits, 3. External device activity, and 4. Blocked email transmissions. Also integrates Big Five Psychometric scores (Openness, etc.).	Architecture: A hybrid "Complex of Algorithms" (CA). It utilizes XGBoost (Gradient Boosting) for tabular classification. The automated part finds latent patterns in the feature-set, which are then "intersected" or "unioned" with a rule-based expert system to eliminate False Positives.	
9	Al-Shehari T and Alsawail (2021) One-hot Endcoding	Dataset: CERT R4.2. Steps: Data aggregation across 5 log types; One-Hot Encoding to convert categorical text to binary vectors; Unix Epoch Time conversion for timestamps; and SMOTE to balance the high disparity (346 malicious vs. 2.3M normal instances).	Indicators: Targeted indicators for "Quitter" scenarios: 1. Removable drive usage by users who previously didn't use them, 2. After-hours login events, and 3. Exfiltration-related URL access (Wikileaks).	Architecture: Comparative analysis of Supervised Classifiers . The study identifies that Decision Trees (DT) and Random Forest (RF) are most robust under adversarial conditions. The use of One-Hot encoding allows the models to learn latent categorical relationships without assuming an artificial numerical order.	
10	Al-shehri Abdullah (2022)	Dataset: CERT (32M events). Steps: Represents audit data as	Indicators: Focuses on inter-relation dependencies . Instead of	Architecture: Rel-RNN (LSTM-based) . It maps embeddings to a Direct	

		Multivariate Time Series. Each activity is an ordered set of real values. Uses Embedding Vectors ($\vec{v}$) to capture characteristics in a multidimensional manner before training.	static rules, it uses a graph-based approach where one activity (e.g., login) is used to model the behavior of another (e.g., HTTP access) through similarity scores.	Graph Structure ($V \rightarrow (N, E)$). The LSTM memory cells capture long-term temporal dependencies, while a dot-product similarity measure identifies latent relations between multivariate sequences to detect displacement in behavior.	
11	Al-Shehari T et.al (2023) Isolation Forest	Dataset: CERT r4.2. Steps: Consolidation of Logon, HTTP, Email, File, and Device logs. Features are converted to numerical vectors. The study specifically evaluates Contamination Ratios (0.00 to 0.1) to find the optimal outlier threshold.	Indicators: Targeted "Quitter" scenario: 1. Increased USB usage after hours, 2. Abnormal logon/logoff times, 3. Specific high-risk URL access (WikiLeaks). It uses a "One-class classification" mindset.	Architecture: Isolation Forest (IF). Unlike traditional ML that profiles "normal," IF exploits the "few and different" nature of insiders. It builds random binary trees; malicious acts are identified by their shorter path lengths to the root, representing high isolation.	
12	Al-Shehari and Alsawail (2023) Random resampling	Dataset: CERT r4.2. Steps: Aggregates 5 log types (HTTP, Logon, Device, File, Email). Converts timestamps to Unix Epoch Time . Solves class imbalance (0.015% malicious) by comparing Random Under-sampling (majority reduction) vs. Random	Indicators: 1. Users who don't normally use a device beginning to use one; 2. Users logging in after work hours; 3. Employees leaving the organization soon after these incidents. Features: # of after-hour logins, # of USB usages, # of Wikileaks visits.	Architecture: Comparative Supervised Learning with Resampling. The "representation" is improved by addressing the skewness in the feature space. Identifies that Decision Trees (DT) combined with Under-sampling achieve the best separability (ROC-AUC) by establishing more	

		Over-sampling (minority duplication) vs. Hybrid methods.		robust decision boundaries in distorted data.	
13	Al-Shehari et al (2024) comparative evaluation + CNN	Dataset: CERT r4.2. Steps: Addresses the extreme sparsity of insider samples. Compares SMOTE (linear interpolation), Borderline-SMOTE (sampling near decision boundaries), and ADASYN (density-based sampling).	Indicators: Classifies threats into Malicious , Negligent , and Infiltrators . Uses these behavioral categories to evaluate how balancing techniques affect the detection of different threat types.	Architecture: CNN (Convolutional Neural Network). The model treats behavioral logs as grid-like structures where the CNN extract local spatial-temporal correlations. ADASYN is found to be the superior "automated" balancer, as it focuses on generating data in regions where the minority class density is low.	
14	Al-Shehari et al (2024) LOF	Dataset: CERT r4.2. Steps: Implements Min-Max Scaling to standardize features $[0, 1]$ and One-Hot Encoding for categorical IT logs. Focuses on Algorithm-level balancing (outlier isolation) rather than data-level rebalancing.	Indicators: Number of logins outside business hours, USB device connection counts, and WikiLeaks URL visits. Uses these to define the "Outlier" status in a human-centric context.	Architecture: Density-Based Local Outlier Factor (DBLOF). Unlike distance-based models, DBLOF calculates Local Reachability Density (LRD) . It identifies threats by finding points with significantly lower density than their neighbors, representing anomalous deviations in behavior sequences.	
15	AlSlaiman (2023)	Dataset: CERT r4.2. Steps: Employs Gray Encoding to represent categorical data,	Indicators: Integrates Sentiment Analysis to categorize the "mood" or intent behind user activities	Architecture: LSTM (Long Short-Term Memory). The model is trained on reformed variable-length sequences. It	

		specifically to ensure that small changes in behavioral sequences result in small changes in the binary representation, preserving temporal proximity.	(e.g., email content or web searches) before the DL phase.	compares Gray Encoding against Binary Encoding (BE) and Real-Valued data, finding that Gray Encoding is superior for maintaining the temporal logic of correlated activities.	
16	Amiri-Zarandi (2023)	Dataset: CERT r4.2. Steps: Processes logs into daily user summaries. Uses Federated Learning to keep data localized on "owner" nodes, sharing only model weights to protect privacy while increasing the training volume through collaboration.	Indicators: Leverages SHAP (SHapley Additive exPlanations) to provide feature-level transparency. Instead of pre-set rules, the model explains <i>after-the-fact</i> which behaviors (e.g., specific file access) triggered the alert.	Architecture: Deep Autoencoder + GRU. The Autoencoder learns a compressed representation of "normal" behavior. The GRU (Gated Recurrent Unit) is used to capture temporal dependencies in sequence data, specifically tuned for the resource-constrained environments of IoT.	
17	Amuda (2022)	Dataset: CERT v4.2. Steps: Integrates logon, file, and device logs. Uses Mutual Information Gain (Algorithm 2) to rank feature relevance (finding 'date_hour' as the 1st rank) and Min-Max Scaling to normalize features to a [0, 1] range.	Indicators: Tracks 10 specific encoded actions including logon/logoff, file copies (by extension: .exe, .doc, .pdf, .txt, .jpg, .zip), and device connect/disconnect. Targets "After-hour" logs (5pm-8am).	Architecture: Hybrid CNN-GRU. The CNN module extracts local features and spatial patterns from sequence data (vector matrices), while the GRU captures long-term temporal dependencies through memory cells and reset gates.	
18	Anakath (2022)	Dataset: Open-source behavioral datasets evaluated in a Cooja	Indicators: Mouse Dynamics: Dragging, double-clicking, movement	Architecture: Deep Belief Neural Network (DBN). The model uses stacked	

		cloud simulator. Steps: Categorizes interaction into "Rich Keystroke" and "Rich Mouse" operations (range [0, 1]). Uses the N-grams method and Jaccard coefficient to calculate distances between behavioral lists.	angle, and speed. Keystroke Dynamics: Key event duration (press/release) and latency (interval between keys) for specific groups (KAlpha, KFunc, KCtrl).	Restricted Boltzmann Machines (RBM). It learns via a visible-to-hidden layer "Positive Phase" and a hidden-to-visible "Negative Phase," optimizing weights through a sigmoid activation function to identify abnormal density in user interactions.	
19	Anju (2024)	Dataset: CERT R4.2 and R6.2. Steps: Employs Z-score normalization to define upper/lower bounds for outlier removal. Uses Mean Imputation for missing sensor data. Features are encoded into integer sequences (Day 0-6, Time 1-24, Activity 0-7).	Indicators: Targeted indicators based on scenarios: 1. Data exfiltration via USB after office hours; 2. Surfing job sites; 3. Installing keyloggers; 4. Login to different employer accounts. Actions include: file access, mail composition, and HTTP processing.	Architecture: Stacked CNN-Attentional BiGRU. CNN (VGG19 & Xception) extracts temporal and semantic discriminative features. Attentional BiGRU processes these as multivariate time-series to model long-range dependencies. Hyperparameters are tuned using Modified Equilibrium Optimization (M-EO) with chaos-based strategies.	
20	Asha (2024)	Dataset: CERT. Steps: 1. Data Aggregation: Feature concatenation of logon, device, and HTTP tables. 2. Transformation: Categorical encoding for text and Epoch conversion for dates. 3. Sampling:	Indicators: Monitors user activity vectors: Logon/Logoff events, Device connections, and HTTP browsing history. Features include User ID, PC ID, and specific Activity types.	Architecture: Double-Layer Anomaly Detection. Layer 1 utilizes an SVM classifier to evaluate and select the best sampling technique. Layer 2 applies the sampled data to One-class SVM and five other anomaly detection	

		Identifies Nearmiss2 (NM-2) as the optimal undersampling approach for balancing the dataset.		techniques to isolate malicious outliers based on density deviations.	
21	Budžys (2024)	Datasets: CMU and GREYC-NISLAB (30,000+ samples). Steps: 1. Outlier Removal: Uses Z-score filtering to remove abnormal typing instances. 2. GFMT: Applies Gabor Filter Matrix Transformation to turn 31-dimension numerical vectors into 2D behavioral images.	Indicators: Keystroke Dynamics: Focuses on the "Password typing rhythm." Metrics include: HT (key duration), KPT (press-to-press interval), and KIT (release-to-press interval).	Architecture: Siamese Neural Network (SNN). The architecture uses two identical CNN branches that share weights. It extracts feature vectors from GFMT images and calculates the Euclidean distance between them to verify identity.	
22	Cai (2024)	Datasets: CERT r4.2 and r5.2. Steps: Integrates multi-source logs; splits sequences into sessions; maps timestamps into 24 hour-based time slots. Utilizes a Hierarchical Navigable Small World (HNSW) index for efficient neighbor retrieval from an activity vector pool.	Indicators: Targeted attack behaviors: Data exfiltration, Data deletion, Compromised passwords, and Modification of critical data. Activity-level units: open file, connect device, logon/logoff, etc.	Architecture: LAN (Learning Adaptive Neighbors). Uses a Sequence Encoder (LSTM/GRU/Transformer) for temporal dependencies. Graph Structure Learning adaptively constructs a local activity graph to learn relationships across sequences. A GNN (GCN/GAT) aggregates these adaptive neighbors, and a Hybrid Prediction Loss mitigates class imbalance by integrating	

				self-supervision and supervision signals.	
23	Ding (2024)	Dataset: CERT v4.2. Steps: 1. Associated Session Graph: defined by 7 rules to link intra-session and inter-session logs. 2. Graph Normalization: Node elimination technique to remove non-critical boundary/session nodes and merge features. 3. Data Augmentation: Synthesizes anomalous samples by inserting benign activities into known attack patterns (Algorithm 2).	Indicators: Rule 2 (Logon): Flags logins outside 8 AM–5 PM or on weekends. Rule 3 (Keywords): Blacklists URLs/filenames (e.g., wikileaks.org, job sites). Rule 7 (Stats): Tracks the sum of malicious HTTP and device actions per session. Scenarios: AB-I (Exfiltration), AB-II (Job search + theft), AB-III (Keylogging).	Architecture: ASG-ITD. Uses a Graph Convolutional Network (GCN). It learns feature representations from a normalized heterogeneous graph containing Core (critical activity), Boundary (login/logoff), and Session (virtual aggregator) nodes. It uses a Graph Laplacian Regularization loss to maintain structural smoothness while detecting outliers.	
24	Dong (2025)	Dataset: CERT r5.2 and r6.2. Steps: Employs TCN for feature fusion. Uses dilated causal convolutions (window size W_c) to capture multiscale temporal dependencies without computational overhead. This creates context-infused embeddings for normal behavior.	Indicators: Traditional CERT vectors (Logon, HTTP, Email, File, Device). Focuses on scarcity of malicious events and high-dimensional sparsity as the primary indicators for anomaly scoring.	Architecture: DDCC (Denoising Diffusion with Curriculum Control). 1. DDPM: A forward diffusion process adds Gaussian noise, and a reverse process (using ResNet) reconstructs the sequence. 2. CL: A soft-weighting curriculum adjustment (h) organizes training from "easy" (common normal patterns) to "hard"	

				(complex/ambiguous sequences) to prevent overfitting on noise.	
25	Dosh (2021)	Dataset: CERT R4.1 and R4.2. Steps: Aggregates logs by User ID. Performs Categorical Encoding to transform text strings into numeric data. Applies Min-Max Normalization to map values into a specific scale [new_min, new_max] to prevent features with large values from dominating.	Indicators: Manual selection of five high-relevance features: Date, User, Source (PC), Action (Activity), and Vector. The study notes that these were selected based on external technical guidance (Towards Data Science).	Architecture: Traditional Machine Learning. Uses Random Forest (ensemble of uncorrelated decision trees), Naive Bayes (probabilistic classifier based on Bayes theory), and 1-Nearest Neighbor (lazy learner using Euclidean Distance to classify instances based on the nearest neighbor in the feature space).	
26	Duan (2024)	Dataset: CMU-CERT r4.2. Steps: 1. Content Scaling: Uses NLP to define maliciousness scores for emails, web pages, and files. 2. SAE Encoding: A Sparse Autoencoder (SAE) with a 1024-node hidden layer encodes 1D behavioral data into 32x32 grayscale images. 3. Normalization: One-hot encoding for roles and Z-score normalization for behavior.	Indicators: Psychological Features: Five major personality traits (Big Five scores). Role Features: LDAP one-hot encoded employee roles. Behavioral Features: Maliciousness levels of accessed content and standard activity vectors (HTTP, Email, File, etc.).	Architecture: Attention-ResNet Fusion. An Attention module captures contextual feature relations to generate high-confidence images. These images are processed via a pre-trained ResNet. Finally, a Multi-source Feature Fusion algorithm unifies behavioral images with psychological and role vectors for final classification.	

27	Erola (2022)	Dataset: Real-world enterprise logs (proxy, VPN, AD, file access) from three distinct corporate environments. Steps: Data is ingested via a modular architecture, normalized into a unified schema, and enriched with organizational context (e.g., job roles and access levels) before being fed into detection engines.	Indicators: Focuses on "Out-of-Character" behavior. Indicators include: Accessing sensitive files outside the usual scope, VPN connections from unusual locations or times, and atypical data transfer volumes via proxy servers.	Architecture: CITD (Corporate Insider Threat Detection) Framework. A hybrid engine that combines Statistical Profiling (to establish a baseline per user) with Unsupervised Machine Learning (e.g., Clustering and Isolation Forests) to detect deviations. It uses a "consensus" mechanism to aggregate scores across different data sources.	
28	Fei et al., (2023)	Datasets: CERT r4.2 (32,770,227 activities) and LANL (1B+ logs). Preprocessing: NLP pipeline (syntax removal, stop-word removal via NLTK). Textual FE: Used 50-dimensional pre-trained GloVe word vectors. Text representation calculated as $V_{\{c\}} = \sum V_{\{w_i\}}$. Temporal: Log sorting and sessionization (logon to logoff).	Emotion Dictionary: Used HowNet to calculate a "negative degree" score for inter-employee dialogs. Contextual Indicators: Defined three behavioral patterns: Attention Anomaly (AA), Emotion Anomaly (EA), and Behavior Anomaly (BA). Ancillary Factors: Developed login anomaly factors ($LA_{\{user\}}$, $LA_{\{host\}}$) to measure the rarity of employee-host relationships.	Fusion Strategy: Log embeddings generated by fusing textual information (AA/EA results) as "ancillary features" with statistical operational features. Detection: Utilized an unsupervised Isolation Forest-based framework for outlier detection at both user and department levels.	
29	Feng (2025)	Dataset: CERT R4.2. Steps: 1. Data Fusion:	Indicators: 70 specific anomalous users. Key	Architecture: MG-UABD. 1. Coarse-Grained (CAD): A	

		Integrates network, login, file, email, and USB logs. 2. Chronological Splitting: Training set = first 25 weeks; Testing set = remaining weeks (Extrapolation mode). 3. Sampling: Hybrid SMOTE + Random Undersampling to balance the 0.02% anomaly ratio.	behaviors tracked include copying specific file types (.exe, .zip), visiting hacktivist/job-hunting sites, and external email recipients. Focuses on the "Pilot Experiment" finding that 90% of anomalous users exhibit <2% anomalous behavior.	Random Forest (RF) model screens all activities to flag potential anomalous users (prioritizes recall). 2. Fine-Grained (FAD): Individualized RF models are built for flagged users to learn personalized behavior patterns (prioritizes precision).	
30	Ferraro (2025)	Datasets: PicoDomain (Zeek logs: connection, Kerberos, authentication, etc.) and CERT r5.2. Steps: 1. Activity ID aggregation. 2. Category partitioning (\$S_{k,j}\$). 3. Prompt-based ingestion with JSON validation to mitigate LLM "hallucinations."	Indicators: Unusual connection duration, byte/packet counts, failed Kerberos authentication attempts, and specific signs of exfiltration/reconnaissance. Uses Chain-of-Thought (CoT) to generate human-readable justifications (Assessment, Reasoning, Indicators).	Architecture: Hierarchical GABM. 1. Specialized Agents: Domain-specific experts for different log types. 2. Supervisor Agent: Aggregates reports for a final holistic classification (\$F_k\$). Relies on the reasoning capabilities of LLMs rather than traditional black-box deep learning.	
31	Gayathri (2024) SPCAGAN	Datasets: CERT r4.2 and r5.2. Steps: 1. Feature Engineering: Extracts context-based features across 5 resource categories (Logon, Email, Web, File, Device). 2. Refinement: Adds sentiment scores and	Indicators: Data volume spikes to USB/external drives, confidential material transfers to personal accounts, abnormal access timing, and unlawful/unethical sentiment in text logs.	Architecture: SPCAGAN + BNN. 1. SPCAGAN: A GAN that uses linear manifold learning to prevent "mode collapse" and ensure high-quality synthetic samples. 2. Bayesian Neural Network (BNN): A probabilistic hybrid model that handles	

		holiday details. 3. Dimensionality: Uses Principal Component Analysis (PCA) to estimate the intrinsic dimensionality (ID) of the data manifold.	Integrates Big Five psychometric scores.	uncertainty better than deterministic DL, increasing resistance to adversarial attacks.	
32	Gayathri R (2025) Adversarial Training	Datasets: CERT v4.2 & v5.2. Steps: 1. Trigger Generation: Uses SHAP to identify positive-contribution features for "malicious" classification to design a stealthy backdoor. 2. Poisoning: Injects poisoned samples into the training set (poison-label attack). 3. Adversarial Training: Augments training data with CWGAN-GP synthetic samples to improve robustness.	Indicators: High-ranking SHAP features such as unusual connection states, data exfiltration patterns, and sentiment-based triggers (abusive language in emails). Incorporates psychometric data to identify vulnerable/disgruntled actors.	Architecture: SNN-MLP & SNN-1DCNN. 1. Self-Normalizing Networks: Uses Scaled Exponential Linear Units (SeLU) to maintain activation normalization. 2. TabNet: Implements attentive interpretable tabular learning. 3. Defense: Uses Adversarial Training (AT) as a primary defense, demonstrating that CWGAN-GP is the most resilient against backdoor poisoning.	
33	Gayathri B (2025) ITD-GMJN	Datasets: CERT, NSL-KDD, UNSW-NB15. Steps: 1. Activity splitting (Standalone vs. Sequential). 2. GOA-MICE: Multivariate Imputation by Chained	Human-Centric Features: Focuses on "Subject Expertise" for standalone logs. Specific Indicators: Deviations in daily schedules, specifically	Architecture: LS-AE + JNN. Process: 1. LS-AE: An LSTM-based Autoencoder extracts	

		Equations where parameters are optimized via the Golf Optimization Algorithm to minimize MSE in missing data.	First Logon Time and frequency of unauthorized access attempts in cloud PaaS/IaaS environments.	automatic features from one-hot encoded activity sequences. 2. Jordan Neural Network (JNN) : A recurrent structure where the Context Layer receives feedback directly from the Output Layer to maintain state memory of previous activities.	
34	Gonzales (2025)	Datasets: Synthetic action databases and UniProt protein sequences. Steps: 1. Contiguous Subsequence Extraction . 2. Subpattern Pruning (retaining only maximum patterns). 3. Local Context Weighting (LCW) to adjust probabilities based on immediate action dependencies.	Human-Centric Features: The ABC Model . It identifies the "Antecedent" (trigger/dissatisfaction), "Behavior" (the action sequence), and "Consequence" (the outcome/data theft). It models how the middle behavior (Y) influences the transition between the trigger (X) and outcome (Z).	Architecture: C3P (Contiguous, Contextual, and Classifying Pipeline). Process: It creates a probabilistic global model using bigrams and trigrams . It calculates a Context-Aware Divergence (CAD) score by measuring the difference between observed and expected behavior using KL-Divergence .	
35	Greitzer (2021)	Dataset: SOFIT Ontology + Expert Survey (20 experts).	Human-Centric Indicators: Utilizes the SOFIT taxonomy (~200 features).	Architecture: Hierarchical Bayesian Network (BN). Process:	

		Feature Engineering (FE) Steps: 1. Quantification: Transformed subjective "Level of Concern" (r_i) into numerical Likelihood Ratios (L_i). 2. Non-Linear Scaling: Applied a Quadratic Regression formula: $L_i = 0.0003r_i^2 + 0.0023r_i + 0.0083$ to engineer the feature weights. 3. CPT Engineering: Derived Conditional Probability Tables by mapping L_i to probability space via $p(I_i A) = L_i / (1+L_i)$.	Key Behavioral Indicators: • Psychological: Disgruntlement, ego, narcissism, anger, and "Dark Triad" traits. • Life Narrative: Unexplained affluence, recent divorce, gambling addiction, or being passed over for promotion. • Boundary Violations: Workplace violence and unprofessional conduct.	1. Latent Representation: The model abstracts individual indicators into "Behavior" nodes (parent nodes). 2. Hierarchical Learning: The "Behavior Bayes" model represents features in a hierarchy to account for conditional dependencies, ensuring that overlapping evidence (e.g., two similar aggressive acts) is not double-counted, refining the model's accuracy in predicting expert-labeled ground truth.	
36	Rahman (2022)	Dataset: CERT r4.2. Technical Steps:	Unintentional Indicators: • Access: Hardware, Software, and Network access permissions.	Architecture: NARX Neural Network. Process: A recurrent dynamic network designed for non-linear time-series.	

		1. Partitioning: Evaluated multiple data split ratios (e.g., 70:30, 80:15:5) to find the optimal training/validation balance. 2. Extraction: Activity logs (Logon, Email, Web, USB) were filtered to extract features belonging to the "Insider Level" and "Threats Level" components.	• Human Factors: Policy violations, human error, and inadvertent negligence. • Contextual: Insider Knowledge (familiarity with internal secrets/ICT procedures) and Level of Trust granted to the individual.	It uses a multilayer perceptron with a sigmoid hidden layer and a linear output layer. It incorporates time delays and output feedback loops to simultaneously learn temporal dependencies within sequences and external exogenous signals from user activity logs.	
37	Haq (2022)	Dataset: Enron Email Dataset. Feature Engineering (FE) Steps: 1. Temporal Engineering: Decomposed date strings into granular features (year, month, day, hour, minute, second).	Human-Centric Indicators: • Person of Interest (POI): Labels employees involved in the corporate scandal using combined financial and email metadata. • Semantic Context: Analyzes email content for suspicious intent, unauthorized information sharing, and communication anomalies.	Architecture: Hybrid LSTM + ML Classifiers. Process: 1. Embedding Layer: Word2vec/GloVe captures context and semantic similarity between words using cosine angles. 2. LSTM Layer: A 100-unit LSTM processes the sequence of embedded	

		2. NLP Vectorization: Used pre-trained Word2vec (Google) and GloVe (Stanford) to engineer semantic word embeddings. 3. Data Structuring: Cleaned headers, message bodies, and employee identifiers into a (489236, 9) shape.		vectors to learn temporal dependencies. 3. Ensemble Classification: The XGBoost classifier uses engineered features to identify threats with high precision (0.92) and recall (1.00).	
38	He Daojing (2022)	Datasets: Enron, monkey.org, and CMU-CERT r6.2. Technical Steps: 1. MDFTS Extraction: Gathers multi-domain features from heterogeneous sources. 2. Text Vectorization: Converts email subject and body into numerical vectors for deep learning layers. 3. Fusion: Combines simulation data with real	Human-Centric Indicators: • Psychological Weaknesses: Curiosity, greed, reciprocity, and trust. • Personality Traits: Uses the OCEAN model (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) to score predispositions. • Behavioral Scenarios: Abnormal logon, data theft, and unauthorized mobile device connection.	Architecture: Double-Layer Hybrid Pipeline. Layer 1 (L-XGB): Combines LSTM (for subjects) and Bi-LSTM (for bodies) with an XGBoost classifier for rapid phishing detection. Layer 2 (BiLA-ITD): Utilizes Bidirectional LSTM to maintain long-range temporal dependencies and an Attention Mechanism to highlight key anomalous behaviors in a sequence.	

		logs for platform verification.			
39	He Daojing (2024)	Datasets: Enron (Normal), monkey.org (Phishing), CERT r6.2 (Insider). Feature Engineering: MDFTS Algorithm: Fuses multi-domain data into a 7-tuple: \$(id, userid, date, activity, content, ldap, psychometric)\$. Custom Loss Function: Engineered a log-loss function in XGBoost that penalizes False Negatives (FN) more heavily to prevent phishing leaks.	Sociotechnical Indicators: • Psychometric: Uses the OCEAN (Big Five) model to assess traits like Neuroticism and Extraversion. • Email Features: Encoding types, HTML iFrames, inducement words, and CC information. • Group Dynamics: Analyzes the Attack Chain correlation between external attackers and internal employees.	Architecture: Double-Layer Hybrid DL. Layer 1 (L-XGB): Uses LSTM for short text (Subject) and Bi-LSTM for long text (Body) to feed an XGBoost classifier. Layer 2 (BiLA-ITD): A Bi-LSTM + Attention model that learns long-range dependencies in daily behavior sequences $[w_1, w_2, \dots, w_T]$.	
40	Hong (2023)	Dataset: CERT 4.2. Feature Engineering (FE) Solution: Manual Selection: Expert-driven extraction	Hybrid Indicators: • Standalone: Logon/logoff times, frequency of specific file accesses. • Sequential: Patterns within a 24-hour window	Architecture: ResHybnet (Residual Hybrid Network). Components: GNN Layer: Captures topological behavior connections.	

		from standalone daily actions. 2. Automatic Extraction: An LSTM Auto-encoder compresses long sequences of daily activities into latent feature vectors. 3. Graph Construction: Nodes are defined as User-Day combinations to capture temporal and social connections.	(e.g., File Open $\rightarrow$ Send $\rightarrow$ Disconnect). • Relational: Organizational hierarchy and peer-group connections extracted via GNN.	2. CNN Layer: Discovers patterns in non-topological contexts. 3. Residual Link: Directly connects node attributes to the CNN to prevent the "downgrading" of individual traits often caused by GNN over-smoothing.	
41	Huang (2025)	Datasets: KDD-UEBA, NSL-UEBA, CIC-UEBA, and KDD-UNBLogs. Technical Steps: 1. Encoding: ColumnTransformer for features and one-hot encoding for labels. 2. Resampling: Rare class sampling to balance malicious instances.	Human-Centric Indicators: • Temporal Patterns: Accessing systems during unusual hours or abnormal frequencies. • Sequential Patterns: Deviations in the order of operations (e.g., bypassing intermediate steps to reach sensitive data). • Contextual Patterns: Shifts in login location,	Architecture: DenseAttDNN (Dense Attention DNN). Process: 1. Dense Connections: Every layer receives concatenated outputs from all previous layers, ensuring maximum feature reuse and mitigating vanishing gradients.	

		3. Dimensionality Reduction: Random Forest to filter irrelevant features. 4. Scaling: 1-D Batch Normalization (BN) for training stability.	device type, or network connection environment. • Attack Vectors: Specific focus on U2R (Privilege Escalation) and R2L (Remote to Local).	2. Attention Layer: A Scaled Dot-Product Attention module dynamically calculates weights for input features, prioritizing critical behavioral anomalies (temporal/sequential) while suppressing noise. 3. Deep Refinement: Seven consecutive fully connected layers abstract and refine the feature-rich extraction matrix into a final classification vector.	
42	Hurst (2022)	Dataset: Real-world UK Hospital EHR logs. Steps: Disaggregation of data into four behavioral groups. Feature extraction involves 12 statistical descriptors per vector (Frequency, Mean, Mode, Std Dev, and Percentiles). Data is normalized using a 70–30 split for binary	Indicators: • Routine-based: Unique clinical activities (e.g., pharmacy orders, assessment form history). • User-based: Practitioner identification and access duration. • Device-based: Identification of specific hardware/device numbers used for access.	Architecture: Supervised Machine Learning (SVM-Radial). Process: The Support Vector Machine (SVM) utilizes a Radial Basis Function (RBF) kernel to calculate an optimal hyperplane in a high-dimensional feature space. It processes aggregated statistical vectors to discriminate between	

		classification (Normal vs. Anomalous).	• Patient-based: Identifiers for which records were accessed.	legitimate clinical workflows and anomalous data access patterns.	
43	Jaiswal (2024)	Datasets: CERT r4.2 and r5.2. Steps: 1. Encoding: Numerical mapping of activities (e.g., logon=0, upload=5). 2. Aggregation: Daily frequency calculation of activities grouped by user-day using an outer join of five log sources. 3. Normalization: Application of Min-Max scaling to improve model convergence.	Indicators: • Temporal Context: Frequency of 30 specific activities per day to uncover patterns hidden in volume. • Log-Specific actions: Logon/Logoff, file access, email transmission, and web browsing activities. • Scenario-based: Focuses on intellectual property theft, data exfiltration, and sabotage.	Architecture: Two-Step Insider Threat Detection (TSITD). Process: 1. Resampling Step: Uses hybrid (SMOTEENN, SMOTETomek), oversampling (ADASYN, SVMSMOTE), or undersampling (ENN) to rebalance the minority threat class. 2. Classification Step: Employs 8 classifiers (ANN, RVFL, ELM, RF, DT, XGB, kNN, SVM) to identify anomalous user-day sequences.	
44	Janjua (2021)	Datasets: Enron and TWOS (Wolf of SUTD).	Indicators: • Linguistic: Frequent terms extracted from	Architecture: Semi-Supervised K-Means-DT.	

		Steps: Extraction of email content, stop word removal, and stemming. Transformation of textual data into numerical vectors using TF-IDF (Term Frequency-Inverse Document Frequency) .	clusters (e.g., specific suspicious corporate jargon). • Contextual: Timestamps, sender/receiver headers, and psychological personality traits from the TWOS dataset.	Process: Unsupervised K-Means clustering is first used to partition unlabeled emails into normal and malicious clusters. Once labeled, a supervised Decision Tree (DT) is trained on the resulting classes to predict malicious intent in new communications.	
45	Jyosthna (2021)	Dataset: CERT r6.1. Steps: 1. Aggregation: Converts raw logs into statistical vectors (count of late hours, login frequency). 2. Normalization: Min-Max scaling to the [0, 1] range. 3. Reduction: Uses PCA to transform 15-dimensional data into a 2D plane.	Indicators: • Role-based: Partitions users by job role (e.g., Chief Engineer) to establish local "peer" baselines. • Activity Frequency: Total counts of thumb drive connectivity and logon sessions on a weekly/monthly basis.	Architecture: N-Median Outlier Detection (NMOD). Process: Calculates an Euclidean distance matrix for specific role groups. It identifies outliers based on a threshold derived from an N-median distance plot , which determines the severity of deviation (High, Medium, Low) for each user relative to their role peers.	

			• Deviation Score: Calculated based on the distance from the role-group median.		
46	Kabir (2023)	Dataset: CERT 4.2. Steps: 1. Timestamp Conversion: Date/Time fields converted to Unix epoch using the formula $\\$NC = (C2 - DATE(1970, 1, 1)) * 86400\\$. 2. Label Encoding: Manual mapping of discrete vectors (HTTP=0, Email=1, Device=2). 3. Normalization: Ensuring all text strings are encoded into integers for the input layer.	Indicators: • Activity Types: Differentiates between HTTP, Email, and Device usage. • Temporal: Uses Unix timestamps to identify the timing of events. • Entity Mapping: User ID and PC mapping to associate actions with specific insiders.	Architecture: Multi-Layer Neural Network (ANN). Process: A sequential Keras model with three hidden dense layers (32, 16, and 8 units) using ReLU activation. It incorporates a Dropout layer (0.5) to prevent overfitting and a final Sigmoid output layer for binary classification (Malicious vs. Benign). Trained with the Adam optimizer.	
47	Kamatchi (2025)	Datasets: Simulated (NS-3) and X-IIoTID. Steps: 1. Clustering: OPTICS-CR is used to group nodes	Indicators: • Session Metrics: Login times and access patterns from IoT devices.	Architecture: Federated Bi-LSTM with FAWO. Process: Local Bi-LSTM models are trained on edge devices to capture past and future sequential behaviors. The FAWO (Federated Automatic Weight	

		and identify cluster heads for localized analysis. 2. Imputation: Mean imputation handles data gaps from intermittent IoT connectivity. 3. Normalization: Scaling of login times and data transmission metrics (bytes) to a common range [0, 1].	• Network Metrics: Data transmission volume (bytes). • Trust Metrics: Digital certificates via hybrid RSA-ECDSA for node registration.	Optimization) method aggregates these updates at the global server, weighting contributions based on local data size and relevance to handle non-IID data drift.	
48	Kan (2023)	Dataset: CERT 4.2. Steps: 1. Temporal Partitioning: Daily activities are split into work hours (07:00–22:00) and off-hours. 2. Feature Cleaning: Deletion of features with constant zero-values (e.g., browsing on others' computers). 3. DA Strategy: A two-level sampling approach that uses clustering to reduce normal samples	Indicators: • Time-Windowed Frequency: Counts of file access, email, logon, HTTP, and device usage segmented by time of day. • Cyber-Actions: Visiting dangerous websites, downloading keyloggers, and abnormal login frequencies.	Architecture: ERPSO-XGBoost. Process: The system uses XGBoost for classification. To solve the parameter-sensitivity issue, it introduces ERPSO (Ethnic Randomized PSO), which adds Gaussian white noise to acceleration coefficients and divides particles into "ethnic groups" to search for the optimal learning rate and tree depth.	

		and SMOTE to synthesize malicious ones.			
49	Kim (2023)	Dataset: CERT Insider Threat Dataset. Steps: 1. Aggregation: 508 detailed features extracted from 5 log files. 2. Signal Decomposition: Applies Discrete Wavelet Transformation (DWT) to decompose user activity logs into high-pass (detail) and low-pass (approximation) components. 3. Denoising: Uses the approximation coefficients to remove "noise" from normal behavior patterns, reconstructing a cleaner learning boundary.	Indicators: • Cyber Activity: Logon/logoff during holidays/off-hours, email attachment size, and number of words in visited URLs. • Entity Context: Organizational hierarchy and user personality traits (OCEAN model from psychometric files). • Statistical: Mean, median, and standard deviation of daily operation frequencies.	Architecture: DWT-enhanced One-Class SVM (OC-SVM). Process: The system treats user behavior logs as a signal. By applying DWT, it compresses the distance between normal data points, making the normal profile more dense. An OC-SVM then identifies malicious actions as outliers that exist outside this tightened normal boundary, significantly reducing the false-alarm rate caused by legitimate but "noisy" behavior.	
50	Kong (2025)	4 Datasets: CERT r4.2, SEA, HCT, PU. Steps: 1. Behavior Mapping: Encodes logs using \$Action ID = Action Factor	Indicators: • UADS (User-Centric): Uses information entropy to measure individual behavior diversity and exploratory tendencies.	Architecture: DPI-ITD (Dual-Perspective Information-Driven). Process:	

		$\times 8 + \text{Dev-Time Offset}$. 2. Sequence Tagging: Computes a Tagging Score (TS) to filter redundant fragments and highlight high-impact behaviors. 3. Session Segmentation: Divides activity into discrete cycles (e.g., daily).	• BCS (Behavior-Centric): Combines global prevalence (γ) and individual frequency (δ) to determine the contextual importance of a fragment. • Tagging Score (TS): $T(u, f) = A(u) \ln[B_u(f) + 1]$, a product of user diversity and behavioral context.	1. Adaptive GloVe Embedding: Dynamically adjusts the co-occurrence window size based on the BCS; rare behaviors get larger windows (l^*) to capture richer context. 2. Detection: Refined embeddings are processed through either an ML model (XGB) or a DL model (Filter-enhanced MLP - FMLP) to distinguish malicious sessions from normal noise.	
51	Kotb (2025)	Dataset: CERT r4.2 (Full). Steps: 1. Relational Mapping: Maps logon, email, device, file, and HTTP logs as entities connected to the user (psychometric) entity. . Deep Feature Synthesis (DFS): Automatically generates	Indicators: • Psychometric Scores: O, C, E, A, and N personality traits. • Relational Aggregates: Total email attachment size, maximum file access frequency per user, and off-hours login counts.	Architecture: DS-IID (Deep Synthesis Insider Intrusion Detection). Process: The model uses DFS to construct a deep tabular representation of users. The classifier is a Binary Deep Learning Neural Network consisting of an input layer followed by 10 identical blocks (Linear +	

		430 features by stacking mathematical primitives (SUM, MEAN, MAX) across relational tables. 3. Balancing: Uses on-the-fly weighted random sampling to address the scarcity of malicious events.	• Adversarial Profiles: Synthetic user profiles generated via CTGAN and Gaussian Copula to mimic legitimate users.	Batch Norm + Leaky ReLU + Dropout). It is specifically trained to differentiate between real normal users, real malicious insiders, and AI-generated synthetic threats.	
52	Lavanya (2024)	Dataset: CERT r6.1 and r6.2. Steps: 1. IPCA: Reduces dimensionality of timestamped user activity without altering the underlying data structure. 2. Robust Scaling: Uses median and interquartile range (Iqr) to normalize data while remaining resistant to outliers. 3. Outlier-Resistant K-means: Groups user functionality samples into 5 clusters corresponding to the benchmark scenarios.	Indicators: • IoT Layer Mapping: Categorizes threats into layers: Perception (device access), Transport (email gateway), Processing (web services), Application (files), and Business (access records). • Scenario-Specific Baselines: Uses the five CERT scenarios (e.g., S1: IT admin, S2: job seeker) as centroids for behavioral clustering.	Architecture: EBiGAN + DNN-PI.	

53	Le Zinc (2021), Exploring anomalous behavior	Datasets: CERT r4.2, r5.2 (Main); r4.1, r5.1, r6.2 (Adaptability). Steps: Auxiliary Reasoning: Uses logic to determine PC-user ownership and standard office hours. Granularity Analysis: Compares user-session (login to logout), user-day, and user-week data types. Feature Types: Extracts frequency (count of actions) and statistical features (mean/std of file/email sizes).	Indicators: Organizational Context: Role, team, and department identifiers. Cyber-Actions: Frequency of off-hour logons, USB connection duration, and file activities on shared PCs. HTTP/Email Details: Size of sent files/emails and visits to job-seeking websites.	Architecture: Hybrid Unsupervised-Supervised Pipeline. Process: Initial Detection: Uses an Autoencoder (AE) or distance-distribution method to flag outliers based on reconstruction error (unsupervised). Feature Engineering: Compares PCA, Random Projection (RP), and the AE bottleneck layer for dimensionality reduction. Generalization: Trains supervised models (Random Forest, ANN, LR) on the limited ground-truth confirmed by analysts to detect unseen insiders.	
54	Le and Zinc-Heywood (2021) Unsupervised ensembles	Datasets: CERT (r4.2, r6.2), LANL, TWOS. Steps:	Indicators: Deviations: "Unusual change" anomalies identified by comparing current activity to a	Architecture: Unsupervised Ensemble Framework. Process:	

		1. Temporal Encoding: Introduces Percentile Representation (\$P_w\$), which converts absolute action counts into a percentile rank relative to the user's own history in a sliding window (7, 30, or 60 days). 2. Feature Extraction: 2-type features (Frequency and Statistical). 3. Aggregation: Evaluates daily and weekly granularity.	personal historical baseline. • High-Risk Actions: Visiting computer monitoring/keylogger sites, downloading files to USB, and logging into supervisors' PCs. • Temporal Shifts: Off-hour activities (e.g., 10 PM) even if the action itself (file access) is legitimate.	1. Individual Detectors: Runs four algorithms in parallel: Autoencoder (reconstruction error), Isolation Forest (tree isolation), LODA (online random projections), and LOF (local density). 2. Ensemble Fusion: Combines scores via Voting (\$VOTE_Inu\$) or Averaging. 3. Robustness: The voting mechanism (specifically \$VOTE_2\$) filters out errors from poisoned training data, ensuring the model identifies malicious users even if the training set is "polluted" with unseen attacks.	
55	Li Dongyang (2021) Image-Based	Datasets: CERT r4.2 and r6.2. Steps: 1. Domain Isolation: Aggregates logs (Logon,	Indicators: • Contextual Proximity: Activity during "After-hours" or "Weekends" vs. standard "Workhours."	Architecture: IGT (Image-based Geometric Transformation). Process:	

		Email, HTTP, File, Device) at three granularities: Week, Day, and Session. 2. Image Synthesis: Converts feature vectors into grayscale images to capture inter-feature spatial relationships. 3. Self-Labeling: Applies 18 geometric transformations (rotations/translations) to normal images to create a multi-class dataset.	• Access Deviations: Logins on "Other's PC" vs. "Own PC." • Specific Intent: Visits to "Job-seeking" or "Leakage/WikiLeaks" websites; execution of ".exe" or ".doc" file operations. • Exfiltration Markers: Mean size of email attachments and frequency of USB device connections.	1. WRN-10-4: A 10-layer Wide Residual Network with 7 convolutional layers and skip connections. 2. Self-Supervised Logic: The model is trained to predict which geometric transformation was applied to an image. 3. Anomaly Scoring: Anomaly scores are derived from the model's classification confidence across transformations using a Dirichlet distribution scoring function.	
56	Li Ximing (2023) Dual-Domain Graph	Datasets: CERT r4.2 and r6.2. Steps: 1. NLP Feature Extraction: Categorizes web pages (Wikileaks, hacking, job hunting) and emails via symbolic tagging.	Indicators: • Web Context: High attention values for Wikileaks-related and hacking-related web browsing (ID 21-26).	Architecture: DD-GCN (Dual-Domain Graph Convolutional Network). Process: 1. Dual Domains: Operates on a Topology Domain (original links) and a Feature Domain (behavioral similarity).	

		2. Graph Construction: Converts users into nodes. 3. Weighted Feature Similarity: Uses Cosine Similarity to build edges between users with similar behavioral patterns, even if no direct "logon/email" link exists.	• Exfiltration: Email sent outside the organization and average email size. • Physical Indicators: Removable device connections and specific file types (exe, zip). • Temporal Logic: Differentiation between Work hour, off-work hour, and weekend logons.	2. Attention Mechanism: Learns adaptive weights (α) to determine which domain or specific feature is more relevant for a given user's classification. 3. HSIC Constraints: Uses Hilbert-Schmidt Independence Criterion to ensure embeddings from different domains capture diverse, non-redundant information.	
57	Li (2024)	Datasets: CERT (r4.2, r6.2) and UMDWikipedia. Steps: 1. Feature Categorization: Extracts features from logon/off, email, device, file, and HTTP activities. 2. Structural Augmentation: Uses GAE to reconstruct the graph, adding edges based on latent behavioral similarities ("soft" relationships).	Indicators: • Relationship Types: "Hard" relationships (email interactions) vs. "Soft" relationships (similarities in behavioral profiles). • Malicious Markers: High-impact behaviors identified through structural importance weights. • Vandalism Detection: Reverts and edit status in	Architecture: GMFITD (Graph Meta-Learning Framework). Process: 1. Structural Reconstruction: GAE + Attention Mechanism establishes latent user connections. 2. Edge Importance Estimation: Prunes	

		3. Episodic Sampling: Divides data into support and query sets to simulate few-shot learning tasks.	Wikipedia as analogs for system sabotage/integrity threats.	potentially perturbed edges (adversarial noise) using cosine similarity and a layer-wise memory mechanism. 3. Graph Prototype Construction: Computes mean embedding vectors for each class in the support set to act as "class prototypes." 4. Meta-Learner (MAML): Optimizes a global initialization that can adapt to new insider threat tasks with just a few gradient updates.	
58	Lilhore et al. (2025)	Datasets: CIC-IoT 2023, UNSW-NB15. Steps: Mean/Mode imputation; Z-score filtering (± 3); Min-Max scaling [0,1]; SMOTE for class imbalance; Sliding window (30/5) for sequences.	Network-Flow Indicators: Packet rate, payload ratio, average inter-arrival time, and connection density. Temporal: Timestamp extraction (Hour of day/Weekday). <i>Note: Lacks human-centric indicators.</i>	Architecture: Hybrid CNN-LSTM-Transformer + RL. Process: CNN handles spatial features; LSTM handles temporal sequences; Transformer manages global attention; Reinforcement Learning (Q-Learning) adaptively updates weights based on reward/penalty signals.	
59	Liu et al. (2025)	Datasets: CERT-r4.2, CERT-r5.2, DARPA1999.	Human-Centric Indicators: Browsing	Architecture: BERT-based Semantic Similarity Model +	

		Steps: Session segmentation (sequences between 'LogOn' and 'LogOff'); 7:3 chronological split; BERT-based time-behavioral vocabulary representation; Triple-set generation for contrastive learning using a "hard sample modulation factor" to manage data imbalance.	job/competitor websites (motivation); specific sensitive URL visits (e.g., WikiLeaks); night-time file access/uploads; USB exfiltration; email contact patterns; and causal relationships between psychological stress and behavior.	Attack Knowledge Graph (AKG) + Preference Propagation Mechanism. Process: BERT extracts bidirectional semantic context from operation sequences to identify "evidence." The AKG maps interactions between users, evidence, and attack types. Preference propagation (inspired by CTR prediction) iteratively traces attack paths to predict future malicious actions.	
60	Manoharan et al. (2024)	Dataset: CERT r4.2. Steps: Merging five CSV files (Device, Logon, Email, HTTP, File); Mean imputation for missing values; Categorical encoding of IDs, roles, and departments into integers; "User-day" classification (1 if any malicious act occurred that day); Downsampling	Human-Centric Indicators: Temporal (Day of week 0-6, Time 1-24); Organizational context (User Role, Functional Unit, Department); Activity-specific labels: Logon (1), Logoff (2), USB Connect (3), USB Disconnect (4), Email (5), File (6), HTTP/URL (7).	Architecture: Comparative study of 8 Supervised ML algorithms, highlighting Random Forest (RF) and AdB (Adaptive Boosting). Process: The framework processes an encoded feature vector per "user-day." It uses RF to aggregate "votes" from multiple decision trees to classify behavior. A key focus is hyperparameter	

		to address 0.03% minority class imbalance.		tuning (e.g., n_estimators for AdB, n_neighbors for KNN) to optimize detection across imbalanced scenarios.	
61	Matos et al. (2025)	Dataset: Synthetic Healthcare Access Logs. Steps: Stratified 80/20 train-test split; Autoencoder training on "normal" samples only to generate reconstruction error scores; Encoding of Role and Department IDs; Generation of binary flags for rule violations.	Human-Centric Indicators: Off-hours access (temporal); Unauthorized role usage (privilege); External IP usage (geospatial); Cross-department access (organizational); Accessing the "wrong patient pool" (contextual/clinical intent).	Architecture: Hybrid Meta-classifier (Autoencoder + LightGBM). Process: The Autoencoder learns latent representations of "normal" access to find subtle statistical anomalies via reconstruction error. These scores are fed into a Meta-classifier (LightGBM) alongside expert-defined rule flags and contextual metadata to produce a final weighted prediction.	
62	Medvedev et al. (2025)	Datasets: CMU (Killourhy & Maxion), KeyRecs. Steps: 1D-to-2D GAFMAT (Gabor Filter Matrix Transformation); Interpolation-based standardization to resolve variable password lengths; Resizing to 40x40 pixels via bilinear interpolation; Standardizing feature	Human-Centric Indicators: Keystroke hold times (duration a key is pressed); keydown-keydown intervals (latency between presses); keyup-keydown intervals (release-to-press latency); and unique behavioral rhythms associated with specific fixed-text passwords.	Architecture: Siamese Neural Network (SNN) with Triplet Loss and CNN branches. Process: The CNN branches extract deep spatial features from the Gabor-filtered "keystroke images." The Triplet Loss function minimizes the distance between the	

		vectors to fixed dimensionality for SNN input.		claimed user's embedding and their baseline (anchor), while maximizing the distance from impostor embeddings (negative samples), allowing for robust identity verification.	
63	Mehmood et al. (2023)	Dataset: CERT (Customized/Merged files). Steps: Averaging neighboring values for 'size' outliers; filling missing 'File Copy' entries via pattern matching; removal of features like 'employee' and 'file tree'.	Privilege Indicators: Horizontal movement (accessing peers' data) and Vertical movement (gaining root/admin access). Behavioral: After-working-hours login, increased frequency of thumb drive (USB) usage, and specific file actions (File Open, Copy, Delete).	Architecture: Ensemble Learning (Bagging and Boosting). Algorithms: LightGBM, XGBoost, AdaBoost, and Random Forest. Process: LightGBM (leaf-wise growth) is used to capture complex non-linear relationships. The framework uses boosting to convert weak learners into a strong classifier by iteratively adjusting weights for incorrectly classified malicious instances.	
64	Mehnaz & Bertino (2021)	Dataset: Wikipedia file repository logs. Steps: blktrace utility for block layer I/O tracing; online translation of sector numbers to file names	Fine-Grained Indicators: Access size (number of blocks); Relative file segment [0, 1] (position in memory); Multi-level temporal granularity (Month-Week-Day	Architecture: Finite State Automata (FSA) + Serial/Parallel Episode Mining. Process: The system models user tasks as	

		(inodes); extraction of 7-feature vectors (uid, fname, atype, sz, sg, ts, sr); normalization of file segments.	hierarchies); Sequential order of events (episodes).	"Episodes" (correlated sets of file accesses). These are represented as Finite State Automata that advance only when a user's sequence matches their historical rhythm. Unsupervised clustering is used for role-mining to profile new users.	
65	Mladenovic et al. (2024)	Dataset: CERT r4.2. Steps: TF-IDF (Term Frequency-Inverse Document Frequency) used to convert email, HTTP, and file content into numerical vectors; Feature scaling; Hyperparameter optimization using a modified Red Fox Optimization (RFO) metaheuristic.	Sentiment Indicators: Malicious intent expressed in email content; anomalies in communication tone; "Keyword Significance" (words appearing frequently in specific documents but rarely across the collection).	Architecture: Metaheuristic-Optimized Machine Learning (XGBoost & AdaBoost). Process: The framework uses NLP (TF-IDF) to represent the <i>context</i> of behavior. A hybrid Adaptive RFO algorithm automatically tunes the hyperparameters of XGBoost/AdaBoost to find the global optimum for threat detection, reducing the manual labor of model configuration.	
66	Nasir et al. (2021)	Dataset: CERT r4.2. Steps: Aggregation of LDAP, Device, Logon, Email, HTTP, and File logs into a single "Master	Behavioral Indicators: Temporal (Day 0-6, Time); Activity Labels: Logon (1), Logoff (2), USB Connect (3), USB Disconnect (4), Email (5), File Copy (6), HTTP (7); Organizational:	Architecture: LSTM-Autoencoder. Process: The model is an unsupervised architecture that learns to "reconstruct"	

		file"; Mean imputation for missing data; Integer encoding for categorical features; Use of "Flexible Time Windows" (User Sessions) instead of fixed hourly windows.	User_role, Functional_unit, and Department.	normal multivariate time-series user sequences. During detection, it measures the Reconstruction Error (Mean Squared Error); a high error indicates that the current user's behavioral sequence deviates significantly from the "normal" historical baseline.	
67	Nikiforova et al. (2024)	Dataset: Real-world audit logs from multiple business environments (488 users). Steps: Parsing audit records into sequential action logs; anonymizing user and action names; building a transition matrix that computes the probability of sequential action pairs based on Markov chain principles.	Behavioral Indicators: Action sequences (e.g., login/exit, open/close, view_main_menu); transition probabilities between steps; "Suspiciousness coefficients" based on the frequency and typicality of action sequences within a specific user group.	Architecture: K-means Clustering + Markov Chain Modeling. Process: The system uses K-means with the Elbow Method to automatically group users into clusters with similar behavior. For each cluster, a Markov Chain model represents the "expected" behavior of that group. Real-time user sessions are transformed into a graph and compared against the group model; any low-probability transition triggers an anomaly alert.	
68	Pal et al. (2023)	Datasets: CERT v4.2, v5.2, v6.2.	Behavioral Action-IDs: Access to Supervisor's PC	Architecture: Stacked-Ensemble (Stacked-LSTM +	

		Steps: Generation of single-day activity sequences; "Equally-Weighted Random Sampling" (EWRS) which undersamples normal behavior and oversamples each of the 5 threat scenarios equally; One-hot encoding of action sequences into vectors of shape 282x1.	(specific to Scenario S3); Emails containing BCC (Blind Carbon Copy) receivers; File transfers categorized by file type (exe, doc, zip); and temporal markers (After office hours vs. Regular hours).	Stacked-GRU) with Additive Attention Mechanism. Process: Multiple layers of LSTMs/GRUs are stacked to capture deep non-linear dependencies. An Attention Layer assigns weights to "critical activities" (e.g., a file copy in a sequence of 100 actions), ensuring the model focuses on the most suspicious moments of the day.	
69	Patel & Iyer (2025)	Dataset: Landauer UEBA Log Dataset . Steps: Extraction of administrative, authentication, and comment-based features; Binary encoding of quantifiable characteristics (login frequency, time spent on actions) into binary values (\$E_p\$).	Human-Centric Indicators: Recurring sequences of user interaction (e.g., login times, access patterns); Support-based thresholding (\$min_sup\$ and \$min_count\$) to establish baselines of normal behavior.	Architecture: Siamese Deep Neural Network (SiaDNN) . Process: The FP-Growth algorithm compresses transaction information into an FP-tree to mine frequent patterns. The SiaDNN (SCNN + DNN) then compares the embeddings of real-time behavior against these patterns to calculate similarity, identifying deviations as potential insider threats.	

70	Peccatiello et al. (2023)	Dataset: CMU CERT v4.2. Steps: User session definition (logon to logoff); extraction of 7 features; TF-IDF comparison of web pages against a job-site corpus; normalization of "off-hours" activity based on a 7:00 AM – 5:00 PM baseline; simulation of continuous data streams.	Behavioral Indicators: diff_begin_first (latency in first login); device_count_out (USB usage outside work hours); url_blocklist hits; count_dwn_exe_file (executable downloads); tfidf_jobsites (intent-based web browsing).	Architecture: Semi-supervised One-Class Classification (Isolation Forest, Elliptic Envelope, Local Outlier Factor). Process: The framework operates in an Online Phase where samples are classified individually. It features a retraining mechanism that selects high-confidence samples from the stream to update the model, allowing it to adapt to behavioral shifts (concept drift) without requiring a fully labeled dataset.	
71	Pérez-Gomariz et al. (2025)	Dataset: Fortune 500 real-world incident logs. Steps: Parsing Security.evtx for Event ID 4624; cleaning system/service accounts; building a directed graph $G = (V, E)$; path extraction via Algorithm 1 (converting adjacency matrices into machine sequences like	Lateral Movement Indicators: Sequence of hops; Node centrality (identifying domain controllers/servers); Technique-specific logins (WMI, RDP, PsExec, SMB); "Low and Slow" timing patterns.	Architecture: Unsupervised Transformer-based seq2seq Model. Process: The model treats the network path as a "sentence" where machines are "words." By training on thousands of legitimate paths, the Transformer learns the "grammar" of the network. Anomalous lateral movements are identified when the model encounters	

		\$[host_1, host_2, host_3]\$).		a sequence of machine hops that has a low probability compared to learned network dynamics.	
72	Prasad et al. (2024) imbalanced Data resolution	Dataset: CERT v5.2 (693,649 samples). Steps: Data cleaning; handling extreme imbalance (1,307 malicious vs. 692,342 normal) via SMOTE (Synthetic Minority Oversampling Technique) and ADASYN (Adaptive Synthetic Sampling); Feature reduction using PCA (Principal Component Analysis) to manage 800+ features.	Behavioral Indicators: Features extracted from technical logs representing user actions and potential "insider" traits.	Architecture: Ensemble Learning (Stacking and Voting). Process: The model uses PCA to condense information before passing it to an ensemble of Random Forest, AdaBoost, and Decision Trees. The Stacking Classifier creates a meta-model that learns how to best combine the predictions of the individual learners to maximize the detection rate of the minority (malicious) class.	
73	Prasad et al. (2025) Hybrid Machine Learning	Dataset: CERT Dataset. Steps: Standardization via StandardScaler; Outlier removal using Interquartile Range (IQR); Class balancing using SMOTE (k=5), Random Undersampling, and	Behavioral Indicators: Login: Frequency, session length, and off-hour (9 PM–6 AM) access. File: Modification/deletion ratios;	Architecture: Hybrid Generative-Discriminative (DAE/VAE + RF/XGBoost). Process: A Deep Autoencoder (DAE) and Variational Autoencoder (VAE) learn a 16-dimensional latent	

		Edited Nearest Neighbors (ENN).	sensitive directory access (HR/Finance). 3. Email: Volume spikes; external recipient counts; executable attachment presence.	representation of "normal" behavior. These Generative Latent Features are fused with the 50 engineered behavioral features. This enriched feature space is then fed into Random Forest and XGBoost for final threat classification.	
74	Prasad et al. (2025) Deep Learning Techniques	Dataset: CMU CERT. Steps: Data cleaning; Addressing class imbalance using SMOTE; Feature Engineering to split the 830-feature set into 8 behavior-specific sub-datasets.	Behavioral Indicators: Organized into 8 categories: 1. Time: Weekend vs. weekday activity. 2. User: ITAdmin status. 3. Logon: After-hour login counts. 4. USB: Mean file tree length and duration. 5. File: Depth and word counts. 6. Email: External recipients and attachment types (exe, zip).	Architecture: Comparative Ensemble (RF, Adaboost, DT) and Deep Learning (ANN). Process: The study trains specialized models for each behavior type. Random Forest and Decision Trees proved superior for high-dimensional classification (e.g., Email-related at 99.8%), while ANN demonstrated a unique ability to capture complex non-linear relationships in USB (89.7%) and Role-related (87.6%) behaviors.	
75	Qawasmeh & AlQahtani (2025)	Dataset: Synthetic Organizational Dataset (10k records). Steps: Conservative missing	Indicators: 17 behavior types with weights: Login outside hours (4), Unauthorized software (9),	Architecture: Real-Time Hybrid Monitoring & Risk Classification. Process: Continuous network	

		value imputation (set to 1); Outlier adjustment; Data balancing via SMOTEENN (SMOTE + Edited Nearest Neighbors); 70/15/15 data split.	Social engineering (8), Thumb drive insertion (10), Secure printing (6), and Network interaction (7).	monitoring feeds an abnormality identification layer (ML-based). Detected anomalies trigger a Risk Score Calculation (Sum of Weights). Employees are then dynamically profiled and classified into risk levels (Low, Moderate, High).	
76	Qi et al. (2025)	Datasets: CERT r4.2 and LANL. Steps: Construction of Temporal Heterogeneous Subgraphs per time window; PC-centric entity linking (files/USBs link to PCs to reduce graph complexity); Domain-level website mapping; USB aggregation per PC; Subgraph sampling centered on malicious nodes for LANL; Positional Encoding to distinguish time-window sequences.	Indicators: Heterogeneous Degree (HD) to measure interaction complexity; Activity statistics (Frequency, Duration, Size: sum, max, min, mean, variance); Attribute features (File extensions like .exe/.zip, website categories like "cloud services"); Scenarios including Lateral Movement (hopping between PCs) and IT Sabotage (keylogger usage).	Architecture: ATHITD (Attention-based Temporal Heterogeneous Graph Neural Network). Process: The ASTHE encoder uses an attention mechanism to learn spatial relationships and short-term evolution from Temporal Neighbors (the same node in previous windows). A Transformer self-attention layer then aggregates these sequences to learn long-term behavioral shifts. Focal Loss is applied to emphasize rare malicious samples during training.	
77	Racherache et al. (2023)	Dataset: Real-world Small (230k flows) and Large	Indicators: Network behavior "Cyber-Personas" (Developers, C-level, HR);	Architecture: CPID (Cyber-Persona Identification) framework.	

		(4.4M flows) network datasets. Steps: Filtering local IPs/MACs; calculating duration from timestamps; MAC vendor lookup for device typing; SMOTE up-sampling for class balance; Min-Max scaling [0,1]; Fasttext skip-gram for hostname embeddings.	Application categories and descriptions; Hostname visit contexts; Device types (Machine vs. Human operated).	Process: A Machine/Human Segregation module (RF/FFNN) filters out machine noise. The remaining traffic is fed to CNN/LSTM models that learn spatial and temporal patterns from windows of flows. Finally, Isolation Forest and One-class SVM detect threats by flagging deviations from the specific "Persona" profile.	
78	Randive et al. (2023 a) pattern-based approach	Dataset: CMU CERT r4.2. Steps: Extraction of single-day features into a 1D vector; SMOTEENN (SMOTE for synthetic oversampling followed by Edited Nearest Neighbors to clean misclassified instances); Conversion of the 1D balanced vector into grayscale image representations.	Indicators: Statistical and frequency-based single-day features derived from resource access patterns (Logon, File, USB, Email) as listed in Table 4.	Architecture: Wavelet Convolutional Neural Network (WCNN). Process: The model bridges computer vision and ITD by treating behavior as an image. It uses Wavelet Transforms for multi-level decomposition instead of standard pooling. This allows the network to perform spectral analysis (capturing frequency-invariant features) alongside	

				spatial analysis, preventing the loss of subtle behavioral details during downsampling.	
79	Randive (2023b) MWCapsNet	Datasets: CERT r4.2 and r5.2. Steps: Mean imputation for missing values; Min-Max normalization; IGTD framework application: identifies feature correlation via Pearson distance and optimizes pixel placement using Manhattan distance to create a visual "mosaic" of behavior.	Indicators: Four Scenarios: 1. S1 (Theft): Off-hours login + USB/upload. 2. S2 (IP Theft): Job site visits + data transfer. 3. S3 (Sabotage): Keylogger usage + bulk email impersonation. 4. S4 (Exfiltration): Remote file searching + personal email usage.	Architecture: MWCapsNet (Multi-level Wavelet Capsule Network). Process: The model uses Wavelet Decomposition to extract spectral (frequency-domain) and spatial features. Instead of CNN pooling (which loses spatial info), it employs Capsules (vectors representing feature properties like texture/position) and Dynamic Routing by Agreement to learn the hierarchical relationships between user behaviors.	
80	Rauf et al. (2021)	Dataset: CERT Insider Threat Dataset. Steps: Conversion of unstructured log-feeds into structured activity-vectors using temporal	Indicators: Activity vectors derived from system logs; Biological "Sensing" principles applied to monitor protein-like concentrations of user events (e.g., unauthorized	Architecture: Bio-inspired Integrated Framework (TAM + PRM). Process: The Threat Analytics Module (TAM) uses unsupervised machine learning (distance-based clustering) to detect	

		criteria; Preprocessing involves calculating a weighted distance metric to measure the spatial distance between activity-vectors.	access attempts, data volume spikes).	anomalies. These anomalies are fed into the Policy Regulation Module (PRM) , modeled as a Policy Regulation Transition System (PRTS) . The system uses an SMT-based theorem prover (Z3) to autonomously synthesize and verify new access control policies in real-time to thwart ongoing attacks.	
81	Saminathan et al. (2023)	Dataset: Simulated organizational environment logs. Steps: Data collection via "keyboard" and "pynput" Python libraries; removal of null values; Min-Max normalization to scale 82 features into the [0, 1] interval.	Indicators: 1. Keyboard: Timestamps, press/release duration, and total count. 2. Mouse: 2D (x, y) coordinates, click types (left/right/center), and movement across 9 screen quadrants. 3. System: CPU/Memory usage (%), bytes sent/received, and foreground application activity.	Architecture: Unsupervised ANN-Autoencoder. Process: The model compresses 82-dimensional behavior vectors into an 8-node latent representation (bottleneck) through three encoding layers (64-32-16). It then reconstructs the input. Threats are identified by calculating the Mean Squared Error (MSE); if the reconstruction error exceeds a threshold derived from normal baseline behavior, the user is flagged as an intruder.	

82	Bin Sarhan & Altwaijry (2023)	Dataset: CERT r4.2. Steps: One-Hot encoding; Deep Feature Synthesis (DFS) to create 69,738 features; PCA for dimensionality reduction (553 features); SMOTE for class balancing; StandardScaler for normalization.	Indicators: Weekday_Logon_After (after-hours activity), Weekday_Logon_Normal, Weekend_Logon, and url_Count (frequency of web visits).	Architecture: Automated Hybrid Feature-ML Pipeline. Process: Uses Deep Feature Synthesis (DFS) to automatically derive relational features from user logs (Entity, Direct, and Relational features). These are processed by an SVM (RBF/Poly kernels), AdaBoost, and a Neural Network (Input-Dense-Dropout-BatchNormalization-Output) to identify data theft patterns.	
83	Senevirathna et al. (2025)	Dataset: CERT Insider Threat Identification Dataset & CCTV Video Data. Steps: Mean/median imputation; backward/forward filling for sequential logs; One-Hot/Rapid coding for categorical factors; frame extraction and spatial normalization (224x224) for video data.	Indicators: Logon patterns (off-hours/unexpected sites), external device/USB connections, large file download sequences, psychometric traits via OCEAN model, and graph-based interaction patterns.	Architecture: CNN-Random Forest Ensemble (Cyber) & Hybrid CNN-LSTM (Physical). Process: The CNN extracts sequential behavioral patterns from logs, while Random Forest manages mixed categorical/continuous data. For physical security, MobileNetV2 extracts spatial frame features, and LSTM captures the temporal evolution of actions.	

84	Singh et al. (2022) Multi Fuzzy Classifier	Dataset: CMU-CERT v4.2. Steps: Missing value imputation (numerical with '0', categorical with 'Missing'); Decimal Scale & Z-Score Normalization; Redundancy reduction via high correlation filters; Isometric Feature Mapping (ISOMAP) to map high-dimensional data to low-dimensional space while preserving global geometry.	Indicators: Logon/logoff and device connect/disconnect timestamps; Frequency of URL visits, email attachments, and file accesses; OCEAN Psychometric Traits (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism); Content-based keywords (career openings, keyloggers, etc.) from files, emails, and HTTP logs.	Architecture: Multi Fuzzy Classifier (MFC) using multiple parallel fuzzy inference engines. Process: First, ISOMAP extracts non-linear behavioral features. Then, TF-IDF scores evaluate keyword relevance. The Emperor Penguin Algorithm (EPA) optimizes feature selection by simulating huddling behavior. Finally, the MFC processes features in parallel through multiple inference engines using Mamdani (if-then) rules, providing membership values [0, 1] for malicious intent rather than a strict binary classification.	
85	Singh et al. (2023) Hybrid Learning Approach	Dataset: CMU-CERT r4.2. Steps: Missing value imputation; Decimal scaling and Z-score normalization; formation of structured activity	Indicators: Time-based features (logon, logoff, file access, and device interaction timestamps); Behavioral drift (deviations from a user's established baseline pattern).	Architecture: Hybrid Bi-LSTM + ANN + SVM. Process: Bi-directional LSTM (Bi-LSTM) extracts temporal features by processing sequences in	

		vectors from raw log-feeds.		both forward and backward directions. A Feed-forward ANN uses distance measurements to select the most relevant features. Finally, a Support Vector Machine (SVM) with an optimized kernel performs the binary classification of users.	
86	Sivakrishna et al. (2025)	Datasets: CMU CERT r4.2, r5.2, r6.2. Steps: Day-wise aggregation; imputation of missing values; numerical encoding of categorical attributes; SMOTEENN (Synthetic Minority Over-sampling + Edited Nearest Neighbors) to balance classes and remove noise.	Indicators: 1. FAR: File Access Ratio (relative to peer group). 2. EDS: Email Deviation Score (statistical deviation from group mean). 3. IAV: Inactive Activity Variance (temporal irregularities). 4. OAF: Odd Activity Fraction (events during abnormal hours). 5. WF: Weekend Flag (binary activity indicator).	Architecture: Bayesian-Optimized XGBoost. Process: The model uses XGBoost to handle tabular behavioral data through gradient boosting on residual errors. Bayesian Optimization (BO) serves as the tuner, utilizing a Gaussian Process surrogate model and acquisition functions (Expected Improvement) to systematically find the optimal hyperparameters that maximize the ROC-AUC score, ensuring robustness against imbalanced distributions.	

87	Song et al. (2024)	Dataset: CMU CERT v6.2. Steps: Preprocessing involves converting log entries into a set of 129 behavioral tokens. Time information is implicitly encoded by mapping statistical vectors to specific positions in a daily sequence (0:00–24:00). The system adaptively selects the time granularity (e.g., 6h vs 8h) that maximizes the covariance of the user's feature sequence.	Indicators: Absolute Time Information (the specific hour of activity); Relative Time Information (the order of events like Logon before File Open); User-Day Behavior Rhythm (the stability of daily working/rest hours).	Architecture: BRITD (Behavior Rhythm Insider Threat Detection). Process: The Stacked BiLSTM layer models the behavioral rhythm by learning from both forward and backward temporal dependencies. An FNN layer in the encoder is specifically added to improve the detection of time-independent behaviors (point threats). The model functions as an autoencoder, identifying threats via high reconstruction error.	
88	Tabassum et al. (2024)	Dataset: Real-world UK Hospital EHR audit logs. Steps: Cleaning of 1M+ records; One-hot encoding for categorical data; Min-Max Normalization (0 to 1); Missing value handling (Mean for numeric, Mode for nominal).	Indicators: User action frequency; Time spent on specific actions; Access date vs. Patient Discharge Date; Specific "Routine IDs" (device/action mapping).	Architecture: Hybrid Unsupervised (IF/LOF) + Supervised (SVM). Process: The Isolation Forest (IForest) builds an ensemble of trees to isolate anomalies (shorter paths = more anomalous). Local Outlier Factor (LOF) measures local density	

				deviation. These unsupervised scores are then used to train classifiers like SVM and Decision Trees , effectively automating the labeling process that previously required expert manual review.	
89	Tao et al. (2023) Temporal convolutional network	Dataset: SapiMouse (2020 release). Steps: Data is segmented into "action fragments" (movement between two points). First-order differences are calculated to capture velocity and direction changes. Outlier removal and min-max normalization are applied.	Indicators: Mouse cursor XY coordinates; Button state (left, right, middle); Event types (move, drag, press, release); dx/dy first-order differences for movement trajectory.	Architecture: ECA-TCN + OCSVM. Process: The Temporal Convolutional Network (TCN) uses dilated convolutions to capture long-term dependencies in mouse sequences without the gradient issues of LSTMs. An Efficient Channel Attention (ECA) module adaptively weights the most important features (e.g., specific velocity patterns). Finally, a One-Class SVM builds a personalized decision boundary for the specific user.	
90	Tao et al. (2025)	Dataset: CMU CERT. Steps: Extraction of time-	Indicators: Activity types (Logon/Logoff,	Architecture: TTT-ECA-ResNet. Process: Employs	

	Test-Time Teaining	series data from raw logs; selection of a 28-day temporal window to capture behavioral patterns; splitting of data into predetermined training/testing ratios.	Connect/Disconnect, File write/delete, Email, HTTP visits) mapped across a monthly time window to detect deviations in routine.	Test-Time Training (TTT) to capture global long-term dependencies with linear complexity. ECA-ResNet18 uses residual connections and channel attention to focus on critical local/short-term behavioral fluctuations.	
91	Tian Zhihong et al. (2024) Dempster-Shafer Theory	Dataset: CERT v6.2. Steps: Preprocessing includes vectorization into a matrix, data cleansing, and dimensionality reduction. Features are aggregated by count into vectors for each distinct source.	Indicators: Behavioral features are grouped into three manual categories: Basic (e.g., logon activity), Content-based (e.g., email/file contents), and Traffic-based (e.g., volume/flow).	Architecture: Attention-LSTM + Dempster-Shafer Theory. Process: Multi-head attention relates different positions in a behavior sequence for representation; the LSTM models "normal" behavior to detect anomalies via prediction error; finally, the D-S Fusion Engine aggregates evidence from multiple channels to update belief scores and confirm threats.	
92	Tian Tian et al. (2025) specific threat scenarios	Dataset: CERT. Steps: Daily behavior serialization; One-hot encoding ; TF-IDF theme extraction from emails/web; Outlier detection via Isolation Forest ; Behavioral vector updating based on	Indicators: Sensitive text themes; email volume/frequency; file creation/deletion patterns; webpage access; punch-in records; and equipment usage outside regular hours.	Architecture: ITDSTS (Transformer-based). Process: Uses a Transformer Encoder with a Multi-head Attention mechanism to compute relationships between any two behaviors in a sequence, effectively	

		abnormal topic/frequency fusion.		capturing long-range temporal dependencies.	
93	Villarreal-Vasquez et al. (2023)	Dataset: Commercial EDR system logs (38.9M events). Steps: Application-based filtering (e.g., Powershell, SVCHost); Feature Transformation \$FT(\cdot)\$; mapping event features to categorical symbols \$E\$ (Vocabulary size: 175); Splitting by machine and time for validation.	Indicators: Actor (Process name); Target (Internal vs. External IP, DLL filenames, Registry keys); Action (Load, Connect); User Type (System-related vs. Non-system-related).	Architecture: LADOHD (3-layer LSTM). Process: Employs an LSTM encoder to learn the structured "language" of system activities. It addresses the OAR problem by using a Dynamic Selection of probable events (\$K\$), where the threshold is determined by the most repeated probability in the model output at each timestep.	
94	Wall & Agrafiotis (2021)	Dataset: CMU CERT r4.1. Steps: Aggregation into daily intervals; workday/non-workday segregation; Z-score based discretization into 7 ordinal intervals; Laplace smoothing to prevent zero probabilities in sparse data.	Indicators: Logons to unusual workstations; Sent email attachment count/size; Removable media file copies (unusual PC); Web access counts; After-hours activity markers.	Architecture: Bayesian Network (BN). Process: Structure is learned via pairwise chi-squared dependency tests. Parameters are defined using Maximum Likelihood Estimation (MLE). Inference is optimized via factorization (Bayes-Ball) to achieve linear time complexity.	
95	Wang Cheng & Zhu (2022)	Dataset: CERT (Organizational System Behaviors).	Indicators: Subject (User ID); Operation Type (e.g., logon, send, visit); Object	Architecture: BIG (Behavioral Identification Graph) + EPC (Event-	

		+ Others Steps: Property discretization; Initial weighting using FastText pre-trained vectors; Weight Smoothing ($S(w_{\{u,v\}})$) to reduce the gap between common and rare associations; Hawkes Process for time-decay modeling; Time-sequential data splitting.	(e.g., website, file); Host (Server/PC ID). These form the nodes of the "Property Graph."	Property Composite) model. Process: Models intra-event (property-level) co-occurrences as entities and relationships in a graph. The EPC model balances fine-grained property associations with coarse-grained event labels to learn multidimensional spatial distributions of user behavior.	
96	Wang & El Saddik (2023) DTITD	Dataset: CERT r4.2 & r6.2. Steps: Tomek Links for majority class reduction; BERT-based contextual word substitution and GPT-2-based context sentence generation for minority up-sampling; Custom numeric tokenization: $x = B * 24 + h$.	Indicators: Work Hours (8am-5pm vs. After-hours); PC Usage (Assigned vs. Other); Removable Media (Directional flow); File Types (.exe vs. others); Website Themes (Job hunting, hacker, etc.); Email Context (Internal vs. External/Competitor).	Architecture: DistilledTrans (Distilled Transformer). Process: Employs a simplified Transformer Encoder stack with a Self-Attention mechanism to capture long-term temporal dependencies in user activity sequences. It omits the decoder to reduce computational cost and prevent information over-processing, while using	

				positional embedding to retain the order of events.	
97	Wang Jiarong, et al. (2023) Deep Clustering	Dataset: CMU-CERT r4.2. Steps: Extraction of five entities (Time [0-23], Host, User, Action, Magnitude); Discretization of action counts; 70% normal training / 30% detection split; Entity embedding (dim 10).	Indicators: Time-of-day (24h clock), Host-ID mapping, Action Magnitude (repetition levels), and context-specific indicators like "WikiLeaks upload" and "Job search website browsing."	Architecture: Deep Clustering Network (Encoder-Decoder + GRU). Process: The GRU-based Encoder maps multi-source sequences into hidden state vectors. A Clustering layer then uses Student's t-distribution to optimize for intra-cluster compactness and inter-cluster separation, identifying anomalies as outliers or sparse clusters.	
98	Wang et al. (2022) Fusion Model	Dataset: CERT r4.2 (HTTP). Steps: Daily grouping (fixed time-span $\Delta t=1$h); Bipartite graph mapping of user-to-URL; Weighting via inverse user frequency; Gaussian kernel distance transformation; Min-max normalization for fusion.	Indicators: Surfing job websites; soliciting employment from competitors; visiting new/unseen domain names as a proxy for malicious resource discovery.	Architecture: Fusion Model (IUAD-PGAD). Process: The individual model (IUAD) uses least-squares approximation on a power distribution to track the unique growth of a user's browsing profile. The peer model (PGAD) constructs a Relationship Network where peer-groups are extracted via a	

				Dominant Set framework to detect behavioral inconsistency among coworkers.	
99	Wang et al. (2024) FeD ITD	Dataset: CERT r6.2. Steps: Data cleaning; domain isolation (Logon, Device, File, Email, HTTP); numeric encoding of behavior; user-day chronological grouping; contextual word/sentence augmentation using BERT/GPT-2.	Indicators: Access time (hourly); activity types (logon/off, USB connect, file write, email send, web visit); psychometric scores (LDAP/psychometric logs).	Architecture: FedITD (Federated Parameter-Efficient Tuning). Process: Employs Pre-trained LLMs (RoBERTa) with LoRA/Adapter layers to reduce communication overhead in a Federated Learning environment. It utilizes Deep CORAL (Correlation Alignment) to perform unsupervised local domain adaptation by aligning feature covariances between source and target clients.	
100	Wei Yichen et al. (2021) Unsupervised Learning	Datasets: CMU-CERT r6.2 and KDD CUP 99. Steps: One-hot embedding of behaviors; padding/truncation to average sequence length; Cascaded Autoencoder (CAE) purification (filtering	Indicators: 164 behaviors derived from a 3-tier matrix (Day-type, Time-slot, Action-type); Meta-data (Role, Team, Supervisor, Psychometric scores).	Architecture: CPJOS (CAEs Purification based Joint Optimization Scheme) + BiLSTM. Process: BiLSTM extracts temporal feature representations from sequences; CAEs purify	

		samples with high reconstruction error).		unlabeled data; a joint network then optimizes dimension reduction and GMM-based density estimation simultaneously.	
101	Wei Zhiyuan et al. (2024) E-Watcher	Dataset: CERT r4.2. Steps: Time-interleaving merging of log sources; user-based data segregation; One-Hot Encoding; and variance threshold-based attribute filtering to remove low-impact data.	Indicators: Shannon Entropy scores for activity significance; 3-σ rule for behavior deviation thresholds; and Impact Ratio (IR) (≥ 0.03) to balance malicious vs. normal samples.	Architecture: Hybrid E-Watcher. Process: Employs Information Gain to establish a personalized statistical profile, intersected with Local Outlier Factor (LOF) to identify outliers based on local reachability and density.	
102	Wen et al. (2023)	Dataset: Enron Email Dataset. Steps: Filtering for rational senders/receivers; Timestamp validation; Weekly aggregation into \$FT\$ (From/To) tuples; and VADER sentiment scoring to generate a unidimensional "EmotionScore."	Indicators: Sentiment polarity (Positive vs. Negative); emotional intensity (compound scores); and the "sociality" of sentiments (how emotions propagate between specific employees).	Architecture: SVD-based Network Analysis. Process: The sentiment communication network is reshaped into matrices and decomposed via Singular Value Decomposition (SVD) into "base networks." Anomaly detection is achieved by monitoring the change in importance of these base networks over time, using Eigenvector	

				Centrality to identify key nodes (employees) involved in the anomaly.	
103	Xiao Junchao et al. (2023) Graph Neural Network	Dataset: CERT r6.2. Steps: Unix timestamp conversion; Factorization of activity types; Word2Vec (wiki-news-300d) for content; Visibility Graph (VG) and Horizontal Visibility Graph (HVG) to convert logs into topological nodes.	Indicators: Time-based edge features (timestamp difference); Distance-based edge features (Euclidean similarity); Attention-based learned weights.	Architecture: MEWRGNN (Multi-Edge Weight Relational GNN). Process: Uses Relational-GCN (RGCNT/RGCND) to process time and distance graphs and CAN-GAT to learn importance weights across edges, providing interpretability via feature contribution ranking.	
104	Xiao Haitao et al. (2024) Unveiling shadows	Dataset: CERT. Steps: Multi-source log integration (User Audit + User Profile); Daily time granularity division; Min-max normalization; One-hot sequence encoding.	Indicators: Statistical: Frequency of login, file open/write; PC Type: Own, Shared, Supervisor, Other; Time: Work/Non-work hours; Psychometric: Openness, Conscientiousness, etc.	Architecture: CATE (Convolutional Attention and Transformer Encoder). Process: A CNN-based subnetwork with Self-Attention captures statistical patterns from an 18x18 feature matrix, while a parallel Transformer Encoder captures long-distance temporal dependencies in behavior sequences.	

105	Xiao Fengrui et al. (2025) SENTINEL	Datasets: LANL, CERT r6.2, TWOS. Steps: Discrete text encoding; Word2Vec for continuous content; BIG Construction (Hosts as nodes, logs as multi-attribute edges); Dissection of logs into short-term Snapshots.	Indicators: Spatio-temporal patterns: (In-Star for server access, Out-Star for scanning, K-Path for lateral movement); Behavioral profiles: (Auth protocols, Logon types).	Architecture: SENTINEL (ST-GNN + EGAT). Process: Edge Graph Attention (EGAT) enhances edge-sensitive behavioral habitual profiles; GCN + GRU captures spatio-temporal dependencies; and a Multi-Timescale Fusion mechanism integrates daily, weekly, and monthly behavioral states.
106	Ye Xiaoyun et al. (2025) Federated Learning	Datasets: CERT v4.2 (1000 users) and v5.2 (2000 users). Steps: Weekly log aggregation; Z-score normalization; DeepInsight conversion (Pairwise cosine similarity -> t-SNE 2D reduction -> 64x64 pixel grid discretization).	Indicators: Psychometric: Big Five personality scores; Organizational: LDAP roles and departments; Statistical: PCA-based feature weighting (top 20 contributing features).	Architecture: Personalized Federated Learning (PFL) + CNN. Process: Clients train local CNN models on behavior-based grayscale images; model updates (not raw data) are aggregated by a central server to build a global model while preserving privacy.
107	Yi & Tian (2024)	Dataset: CERT r4.2. Steps: Normalization (Z-score/Min-Max); Aggregation into daily and	Indicators: Psychometric scores; Organizational roles; Daily/Weekly behavior frequencies;	Architecture: KLI-XG (Hybrid Outlier Representation). Process: Unsupervised algorithms generate outlier

		weekly intervals; Dimensionality reduction of the scoring matrix via Principal Component Analysis (PCA) .	Outlier scores from KNN, LOF, and Isolation Forest.	scores that act as "nonlinear transformations" of raw features. These are fused with the original data to create an augmented feature space for an XGBoost classifier.	
108	Yumlembam et al. (2025)	Datasets: CERT r5.2 & r6.2. Steps: Activity encoding (Type × 24 + Hour); Standard Scaling (Mean 0, SD 1); Session segmentation (max 50 nodes); and Replication-based oversampling to address minority class imbalance.	Indicators: Supervisor PC access; Work-hour deviation; Removable media file transfers; BCC emails to external domains; Flagged URL visits (Wikileaks, Dropbox); and Keylogger-related keyword counts.	Architecture: Dual-Graph GCN + Bi-LSTM. Process: Parallel GCNs extract spatial features from an Explicit Graph (rule-based connections) and an Implicit Graph (Gumbel-softmax similarity). An attention mechanism fuses these graphs, followed by a Bi-LSTM to model temporal dependencies in both forward and backward directions.	
109	Zhang et al. (2021)	Datasets: CERT v4.2 and v6.2. Steps: Aggregation of five log sources by User ID and session; TF-IDF scoring for operated entities (URLs, emails) to capture importance/rarity;	Indicators: Computer ownership (belongs to user); Behavior time; Email recipient/sender rarity; File/URL rarity via TF-IDF; and status flags for Logon, Logoff, and Device connection.	Architecture: Ensemble LSTM with Self-Supervised Regularization. Process: An ensemble of LSTM sub-detectors is trained using an Over-Bootstrap strategy (50/50	

		and avg-TI normalization to (0,1).		malicious/legitimate sampling). A parallel Self-Supervised pretext task (predicting the User ID) forces the LSTM to learn distinctive, user-specific behavior patterns.	
110	Zhu et al. (2024)	Dataset: CERT. Steps: Aggregation of logs into sessions (start: login, end: logout); Feature split into Event Frequency Vector (counts of behavior codes) and Time Feature Vector (timestamps).	Indicators: Logon/Logoff: Working hours vs. late night; File Operations: Copy/Write activities; Device: Removable storage usage; Email: Volume and attachment status.	Architecture: AUTH (TL-AAE). Process: The encoder combines TCN (for local feature correlation) and LSTM (for long-term dependencies). Generative Adversarial Theory (Discriminator) constrains the latent features to a Gaussian prior to reduce reconstruction uncertainty.	